SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)
Assessment Tool Weightage: 30%

Annexure A
CONCEPT MAPPING – ANSWER KEY

1. *Question:* *Complete the concept map: Application → Presentation → ______ → Transport → Network → ______ → Physical, and explain how the missing layers are essential for reliable communication.*

Concept Map:

```
Application → Presentation → Session → Transport → Network → Data Link →
Physical
```

Explanation / Answer:

The two missing layers are the Session Layer (between Presentation and Transport) and the Data Link Layer (between Network and Physical).

Session Layer: It establishes, manages, synchronizes and terminates the communication session (dialog) between two applications. It keeps track of whose turn it is to transmit, and allows recovery from interruptions by resuming from a checkpoint rather than starting over, which supports reliable, organized communication.

Data Link Layer: It converts the raw network-layer packets into frames, adds physical (MAC) addressing, and performs error detection (e.g., CRC) on each link. This ensures that data is delivered correctly across a single physical link before being handed to the Physical Layer for actual signal transmission.

Together, these two layers close the gap between the application's logical data and the physical transmission of bits — the Session Layer manages the conversation, while the Data Link Layer guarantees error-checked, correctly addressed delivery over each physical hop, both of which are essential for overall end-to-end reliability.

2. *Question:* *Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.*

Concept Map:

```
Application/Presentation/Session Layer  →  Data (Message)
Transport Layer                          →  Segment
Network Layer                            →  Packet (adds IP header to
Segment)
Data Link Layer                          →  Frame (adds MAC header/trailer
to Packet)
Physical Layer                           →  Bits (Frame converted to
electrical/optical/radio signals)
```

Explanation / Answer:

As data moves down the OSI stack, each layer encapsulates the unit received from the layer above by adding its own header (and sometimes a trailer), transforming it into a new, layer-specific data unit: Message → Segment → Packet → Frame → Bits.

This transformation supports transmission because each added header carries the control information that layer needs to do its job — e.g., port numbers in the Transport segment, IP addresses in the Network packet, and MAC addresses plus error-check bits in the Data Link frame. Without this stepwise transformation, the raw application data could not be addressed, routed, error-checked, or physically transmitted.

At the receiver, the reverse process (decapsulation) happens layer by layer, stripping each header until the original message is reconstructed for the receiving application, ensuring accurate and reliable communication.

3. *Question:* *Given the concept map: Data Link → MAC Address → Local Delivery; Network → IP Address → Global Delivery. Explain how these relationships ensure complete data delivery across networks.*

Concept Map:

```
Data Link Layer  →  MAC Address  →  Local Delivery (within same LAN/segment)
Network Layer    →  IP Address   →  Global Delivery (across different networks)
```

Explanation / Answer:

The MAC address is a physical, hardware-level address unique to a network interface and is only meaningful within a local network segment. The Data Link Layer uses it to deliver frames directly between devices on the same LAN, hop by hop.

The IP address is a logical address that identifies a device's network and host position globally. The Network Layer uses IP addresses together with routing tables to move packets across multiple interconnected networks toward the correct destination network.

Complete, end-to-end delivery is achieved by combining both: at each hop along the path, the Network Layer uses the destination IP address to decide the next network to forward the packet to, while at every individual hop the Data Link Layer uses the local MAC address (obtained via protocols like ARP) to actually deliver the frame to the next device or router on that segment. So global delivery (Network Layer/IP) is accomplished as a chain of successive local deliveries (Data Link Layer/MAC), ensuring the data ultimately reaches the correct host anywhere in the network.

4. *Question:* *A concept map shows: Error Control → Data Link Layer → Error Detection; Error Control → Transport Layer → Error Correction. Analyze how the interaction between these layers improves reliability in data transmission.*

Concept Map:

```
Error Control → Data Link Layer → Error Detection (CRC/checksum per hop)
Error Control → Transport Layer → Error Correction (ACK/retransmission, end-to-end)
```

Explanation / Answer:

The Data Link Layer performs error detection on each individual link/hop, typically using a CRC or checksum. If a frame is found to be corrupted, it is discarded (or retransmitted locally on some links) before it can propagate further, catching transmission errors close to where they occur.

The Transport Layer performs error correction on an end-to-end basis, using sequence numbers, acknowledgements and timers. If a segment is lost, corrupted beyond local detection, or never acknowledged, the Transport Layer (e.g., TCP) requests retransmission from the original sender, guaranteeing that the complete, correct data ultimately reaches the destination application.

The interaction between the two improves reliability because the Data Link Layer filters out most physical/link-level errors quickly and locally (efficient, low-latency correction), while the Transport Layer acts as the final safety net that guarantees overall data integrity across the entire multi-hop path, including failures the Data Link Layer cannot see (e.g., a lost packet due to congestion or router failure). Together they provide layered, defense-in-depth reliability.

5. *Question:* *Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.*

Concept Map:

```
Application Layer   →  Browser sends HTTP/HTTPS request
Presentation Layer  →  Data formatting / SSL-TLS encryption
Session Layer       →  Session established with the web server
Transport Layer     →  Reliable delivery via TCP segments
Network Layer       →  IP packets routed to server's network
Data Link Layer     →  Frames delivered on local network (MAC addressing)
Physical Layer      →  Bits transmitted over cable / Wi-Fi
```

Explanation / Answer:

In normal web browsing, each layer performs its role in sequence: the browser (Application) creates an HTTP request, which is formatted/encrypted (Presentation), tied to a session with the server (Session), broken into reliable TCP segments (Transport), routed as IP packets (Network), framed with MAC addressing (Data Link), and finally sent as electrical/optical/radio signals (Physical).

Because the layers are interdependent, a failure at any one layer breaks the entire process: if the Physical Layer fails (e.g., cable unplugged or Wi-Fi out of range), no bits reach the destination at all, so nothing above it can function. If the Network Layer fails (e.g., wrong gateway or routing issue), packets cannot leave the local network even though the local link is fine, so the browser request never reaches the server. If the Transport Layer fails (e.g., TCP handshake cannot complete), no reliable session can be formed even if packets are being routed correctly. This shows that OSI layers work as a dependent chain — a fault in a lower layer cascades upward and prevents every higher layer, including the final web page load in the Application Layer, from succeeding.

6. *Question:* *A network issue is represented in the concept map: Physical → Data Link → Network → Transport → Application . Analyze the problem and explain how the concept map helps in troubleshooting.*

Concept Map:

```
Physical    (working)  →  Data Link   (working)  →  Network   (fails)  →
Transport    →  Application
```

Explanation / Answer:

Since the Physical and Data Link layers are working, the device has a good physical connection and can successfully exchange frames on the local network (e.g., link light is up, local ARP/MAC communication succeeds). The failure begins at the Network Layer, which points to a problem such as an incorrect IP address/subnet mask, a missing or wrong default gateway, or a routing/DNS failure that prevents packets from leaving the local network or reaching the destination network.

Because the Transport and Application layers depend on the Network Layer to deliver packets, they fail as a direct consequence — not because of any problem within TCP or the browser/application itself, but purely because the Network Layer cannot deliver the underlying packets.

How the concept map helps troubleshooting: by mapping the status of each layer in sequence, the concept map lets a technician isolate the fault to the exact layer where things stop working, instead of guessing. Here it immediately directs attention to Network Layer configuration (IP settings, gateway, routing) using tools like ping/traceroute, and rules out cabling, NIC, or application-level causes — making diagnosis faster and more systematic (this mirrors the standard OSI-layer / bottom-up troubleshooting approach used in real networks).